Education Center Management System

For

Thaksala- Horana

P.W.A.Nimnadi

July 2024

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July 2024

BIT

UCSC

This dissertation is submitted in partial fulfilment of the requirement of the Degree of Bachelor of Information Technology (external) of the University of Colombo School of Computing

Declaration

I certify that this Final Project Report does not incorporate, without acknowledgement, any material previously submitted for a degree or diploma in any university/institute, and to the best of my knowledge and belief, it does not contain any material previously published or written by another person or myself except where due reference is made in the text.

Signature of Candidate: Date: 20/07 /2024

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Name of Candidate: P.W.A.Nimnadi

Countersigned by:

Signature of Supervisor(s)/Advisor(s): Date: 20/07 /2024

………………………..

Name(s) of Supervisor(s)/Advisor(s): M.P.S.Wijesinghe

Abstract

Thaksala is an educational institute in Horana. It is a private tuition center to teach grade 6 to ordinary level. They have three branches located at Horana, Ingiriya and Bndaragama. Even though the institute already has a student management system they face various problems such as time wastage and resource wastage in both human and material.

Time is wasted during student enrollment process. Specially while entering student details and material issuing. To solve this problem a self-registration feature is introduced to the students. Another problem is material wastage. They keep manual paper-based records for issuing materials, student requests, payment information as precaution to unreliability of existing system. To overcome this situation a reliable system with more features such as student portal to make student requests, feature rich interfaces are provided to enter material issuing details and manage student's payment details is much more required.

Education Center Management (ECMS) is required to developed with Unified Modeling Language. Object Oriented techniques were used to analyze and design the ECMS.CSS3 was used for the styling of the system.HTML5 was used as the markup language. JavaScript, jQuery, AJAX were used for client-side scripting and PHP was used as the server-side scripting. MySQL was used as the DBMS of the ECMS. ECMS is a web-based system developed on MVC architecture and client server architecture. ECMS would enhance the productivity and efficiency of the Education Center.

Acknowledgement

I would like to express my obligation to my supervisor and the teacher Mr.M.P.S.Wijesinghe for providing his valuable direction, new suggestions and comments throughout the project development scenario. I also acknowledge to the University of Colombo School of Computing for supplying this opportunity and directing throughout the project program. And also, I would like to thank the manager, Mr. W.T.S.Perera for giving me this valuable chance to develop a system for the education center management system and for giving his valuable time for this activity.

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List of Acronyms

MVC	- Model View Controller
ECMS	-Education Center Management System
DBMS	- Database Management System
BIT	- Bachelor of Information Technology
OOD	- Object Oriented Design
UML	- Unified Modeling Language
SRC	- Software Requirement Specification

Chapter 1 | Introduction

1.1 Motivation for the Project

Thaksala is an educational institute situated in Horana area. They have three branches located at Millaniya, Bellapitiya and Ingiriya. Even though the institute already has a student management system they face problems such as time wastage and resource wastage both human and material.

Followings are the motivation factors for the project.

- Time wastage

Time is wasted during student enrollment process. Specially while entering student details and material issuing. To solve this problem a self-registration feature is introduced to the students.

- Material wastage

They keep manual paper-based records for issuing materials, student requests, student payment information as precaution to unreliability of existing system. To overcome this situation a reliable system with more features such as student portal to make student requests, feature rich interfaces are provided to enter material issuing details and student payment details is required for giving and efficient service to the students, teachers and parents.

Followings are the ways of time wasting in the education center.

1. During Student enrollment process. Specially while entering student details and material issuing
2. While applying discounts to students.

- Checks previous class registration records of returning students.

Followings are the ways of resource wasting in the institute.

3. Institute keeps file-based records of issuing materials
4. Student requests are recorded in student request book by front desk officer.
    - e.g. Class transfer/leave request, Personal information change request, refund request etc.
5. Institute keeps manual records of student payment information as precaution to unreliability of existing system.
6. Institute's network administrator manually edits database records in some occasions. Such as class transfers, applying late discounts etc.
7. Irregularities in student refund issuing and student promotional discount confirmations.

The institute hopes to develop a CBIS to solve above-mentioned problems. I believe I can develop a web-based system to solve those problems and introduce new business processes to increase efficiency and reliability of the institute.

Also, it would be a great opportunity for me to practice the theories and techniques learned through the entire BIT program in a real development environment and to complete BIT final year project successfully.

1.2 Objectives

Main Goal of this project is to enhance the reliability and efficiency of the business process of the institute by developing a CBIS.

Followings are the objectives of the project to achieve the goal,

- Reduce the typical student and teacher registration process time from 10 minutes to 3 minutes by introducing self-registration feature to the students. & teachers
- Providing much more supportive interface for class registration form to support discount allocations to save time.
- Convert all manual paper-based record system to CBIS. By doing so paper usage can be cut down by 80% ~ 90%.
- Introducing receipt service to resolve problems arising in refund issuing and promotional discount offerings.
- Providing an interface to students to enhance the communication between institute and students as well as student with the teacher.

  Eg: Student requests, their previous transaction details, timetable function

1.3 Scope

Scope is the main outline of the project and it described what areas will be going to covered from the proposed system.

The main scope of this project involves development of web-based student management system.

Followings are the scope of the project,

- Student, Teacher and Parent management
- Student request management
- Payment management
- Class management
- Material inventory management
- User authentication and authorization management

1.4 Structure of the dissertation

This dissertation consists of six main chapters. The phases of development life cycle are described by each chapter.

1.4.1 Introduction

The initialization phase makes the foundation of a software development project, in which some of the most important aspects, such as objectives and scope are identified. That is essential in defining the budget, schedule and outcomes which are the basic concerns of the client.

1.4.2 Analysis

The analysis phase of the development is described by second chapter. Under this chapter overview of the analysis, fact finding techniques, the existing system, functional and non-functional requirements, literature review and the justification for the choice of the development life cycle are described.

1.4.3 Design

The design phase of the development is described by third chapter. Under this chapter comparison of alternative design strategies, justification for the selected design strategy, complete detailed design of the system, data modeling diagrams and the user interface design are mainly discussed.

1.4.4 Implementation

The implementation phase of the development is described by fourth chapter. Under this chapter the tools and techniques used for the implementation, application structure, code explanations and module structure descriptions are mainly discussed.

1.4.5 Testing
The testing and evaluation phase of the development are described by fifth chapter. Under this chapter the description of testing approach, test plan, results of work and the user evaluation are thoroughly discussed.

1.4.6 Conclusion
The last chapter of the dissertation is conclusion. It is included the review of entire development process carried out critical evaluation of the proposed system, the main facts learnt from throughout the project and the new suggestions for future.

Chapter 2 | Analysis

2.1 Introduction

Software development process analysis takes a crucial part. In this phase the existing system is inspected better to identify drawbacks. After that requirements are categorized to eliminate the above issues as well as newly introducing features also are added to the system. In this phase the details of each requirement are captured, explained the scope of the work and how each requirement is going to be fulfilled.

SRS (Software Requirement Specification) is a fully detailed and completed description of the behavior of the software going to be developed.

In this chapter, fact finding techniques, the existing system, functional and non – functional requirements, literature review and the process model are discussed.

2.2 Fact Finding Techniques

In analysis phase, well defined approaches and tools are used in the formal process of information gathering. These techniques are called fact finding techniques.

The popular and most used technique for fact gathering is interviewing. But it is highly time consuming and costly form. By Observation requirements founded by inspecting how people do their own work in real environment in order to identify the scenarios and some unmentioned facts. Examining documentation can be used to obtain a realizing of the office by referring official reports and documents but, it can be varied from the realistic situation and maybe out of the date or old versioned. Questionnaires are most valuable when the people are geographical dispersed.

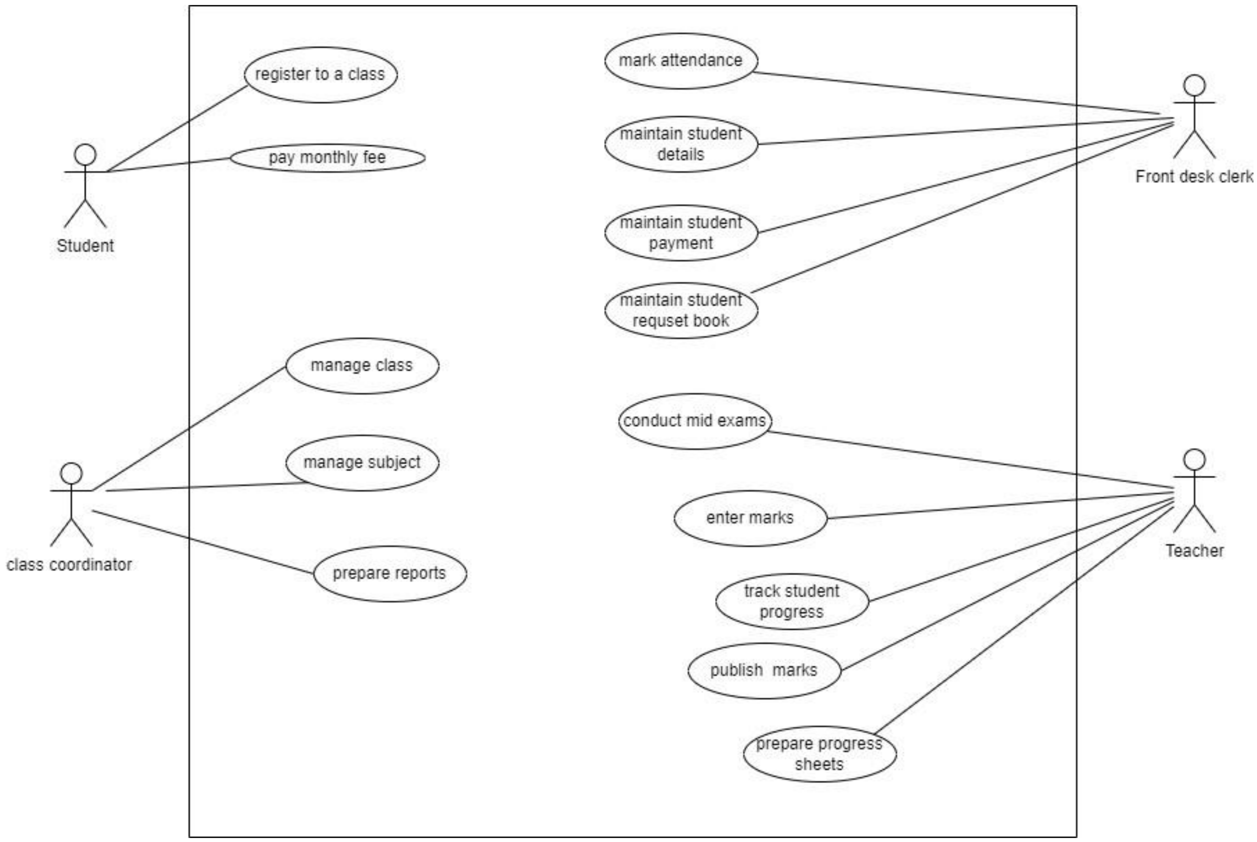
Rather than using one specific method for collecting information, use of multiple techniques for fact finding is very efficient. Interviewing and on-site observation were mainly used for identifying specific requirements for ECMS.

2.3Existing System

Educational Center Management System for Thaksala Horana is currently done manually, which basically consists of a set of books and excel worksheets. The whole system was categorized into a set of sub functional areas as follows for a clear analyzing.

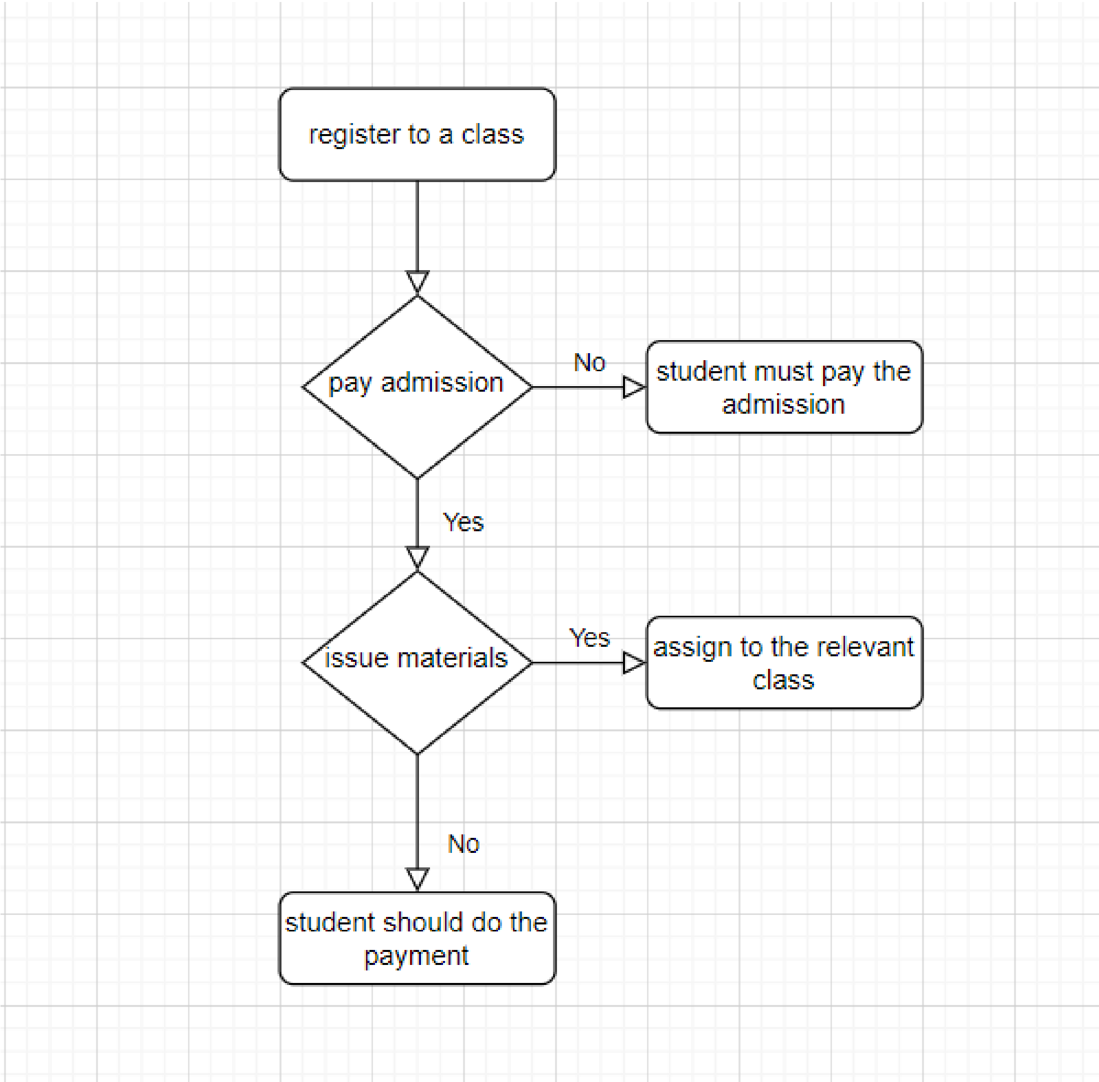
- Student, Teacher and Parent management
- Student request management
- Payment management
- Class management
- Material issuing management
- Report generation

The top-level use case diagram of the existing system of ECMS is given by following figure 2.1



2.1 top level use case digram for existing system

Following figure 2.2 is shown the existing student registration of the system



2.2 flow chart for existing student registration

2.3.1 Student, Teacher and Parent Management

Present system is basically occurred via manually. Students and teachers are registered by giving an admission card. Parents are not much involved to view the progress of students. Teacher has no better facility to inform the students before coming to the physical class to refer their class materials. They use WhatsApp groups to upload class materials. Students haven't a facility to do assignments, quizzes, and pre exams due to the limited time limit before their exams.

Parents are not able to track their children progress in a suitable way. They have to ask from the teachers always. They have not a facility to track attendance of students as well.

Teachers have to wait to the physical class date to give the relevant education material to students. They haven't a suitable way to mark the progress of students and specially in their relevant subject module.

2.3.2 Student request Management

Students have to write their requests in a book when they are needing get a leave from the class or applying to change the class for another branch. As well as when their personal information is changed, they have to write the requests in the book.

When they need a refund, they have to write their requirement in the book. This situation has become a very stubborn for students as well as the administration of the education center.

2.3.3 Payment Management

Students are paid to the class fees at the end of the month. Front desk clerk is entered the paid students' details in excels sheets. Students do not get a receipt when he paid to a specific class.

When a special discount is obtained by student if he is attending to another class in the education center. This calculation can be occurred various incorrections. And also refund issuing is occurring through books and excel sheets. Refund is issued for class fees material issuing etc. After paid the refund for the students they are not able to get refund receipt in current system.

2.3.4 Class Management

Classes are conducting to the grade 6 to grade 11 for various subjects. They have mass classes, group classes and paper classes in the education center. In a particular class room, teacher has to update the students about the required materials that they want to bring to the next date as well as reading materials that students must read. Teachers have not a suitable facility for that.

Teachers have not a facility to measure the progress of students before they face their exams about the contents of the syllabus.

Education center has no facility to inform students about the postponing, rescheduling and cancelling class in a better way.

2.3.5 Material Issuing Management

Materials are issued when a newly registered student come to a particular class. Students are received a past paper book, tutee and other relevant reading materials. issuing is occurred through front desk clerk. They have to maintain a register to issue materials, order new materials as well. But they are unable to search required material quantity after issuing to the students correctly.

2.4 Drawbacks in Existing system

Many and the most common failures in manual systems can be seen through the existing system of this education center management as well. The most crucial drawbacks identified in the system can be mentioned as below.

- It is hard to handle large volumes of data manually. The usual system is consisted of a lot of unnecessary paper works. The details of the students, teachers are gathered via a paper like entrance admission.

- This manual system can be occurred a lot of mistakes, inaccuracies and inconsistent data and information because of the manual procedure. For an example entering discount and payment details accurately is very valuable for the payment management.

- The system has not featured a better way to identify the records uniquely in the system which creates many issues in finding particular records specially in searching data.

- Delays in some processes such as get the relevant tutee because of the need of physical attending. The current system is incapable of providing quickly access to the relevant information as needed time.

- When doing the payments, this situation is very inefficient for payment procedure.

- Summarization of data timely is essential for managerial decision making. But the existing system has been spread through a set of worksheets and files, generating reports can be delayed.

- There is a lack of security level on students related records because most of those are kept on books and unprotected excel sheets in shared computers.

For an example, material issuing details are included in a register which can be accessed by anyone without any trace.

- The current system is centralized to one and the whole process is done through their relevant education center. It has occurred some difficulties to the management to get wide idea about the branch's education capacity.

2.5 Functional Requirements

Functional Requirements are the special features that explains what should be delivered by the proposed system. Following are the highest prioritized functional requirements of ECMS.

- Manage student and teacher details

- ✓ Students can be self-register to the system by web site. They can update their student profile as well as they are able to create students request to change their personal information change, refund request etc.
- ✓ Admin can add teachers to the system. Teachers are assigned to a particular class by class coordinator. Teacher is able to add relevant class materials to the system and he is able to ad quizzes, assignments, pre exams to the system.

- Manage the education center (registering the branch)
  - ✓ Branch manager can be registered the education center.
  - ✓ He is able to view reports and approve teacher commission.
- Manage the class payments, teacher commission
  - ✓ Front desk clerk is done the payment management basically, he is created the teacher commission as well as updated the monthly class payment of the students.
- Manage issuing refunds and discounts to the students
  - ✓ Refunds and discounts are given to the students due to a promotional campaign and populating the education center among students.
- Generate various types of reports

- ✓ Front desk clerk is created various reports about the student attending, payments etc.

- Publish notifications to the students
    - ✓ Notifications are published in the web site about classes as well as cancellation, rescheduling and postponing about the classes.
- Manage users
    - ✓ Users are managed by the admin in the system. They have particular user account, password and user roles to enter to the system.
- Manage the classes and subjects
    - ✓ Class coordinator and the subject teachers are the key components of this module. Teacher create the base plan of the subject module. He is able to mark the progress of the base plan as well as he can make the students progress board.
- Manage material issuing
    - ✓ Materials issuing is occurred through the system when a newly registered student come to a particular class. System will be facilitated to the front desk clerk by searching required material quantity, ordering required materials etc.

2.6 Non-Functional Requirements

The constraints or the restrictions inflicted on the system, which specify the quality attributes of the software are called as non-functional requirements. These are basically the requirements that outline how well the system will operate including them. Non -functional requirements are valuable for satisfaction of user and client expectations. The non -functional requirements of the ECMS have been described below.

- The system must be secure.

Because details of the student's payments will be handled by the proposed system. This confidential information may fall into the hands of outsiders when the system does not provide sufficient security.

- The system should be reliable.

The system should be available at any moment for the users and able to provide valid and accurate data with least failure rate.

- The system should be usable.

The system must not be difficult to use and easy to study, giving even an untrained user to be able to work with it easily and efficiently. It will help to increase the interaction between the user and the interface.

- The system should be portable.

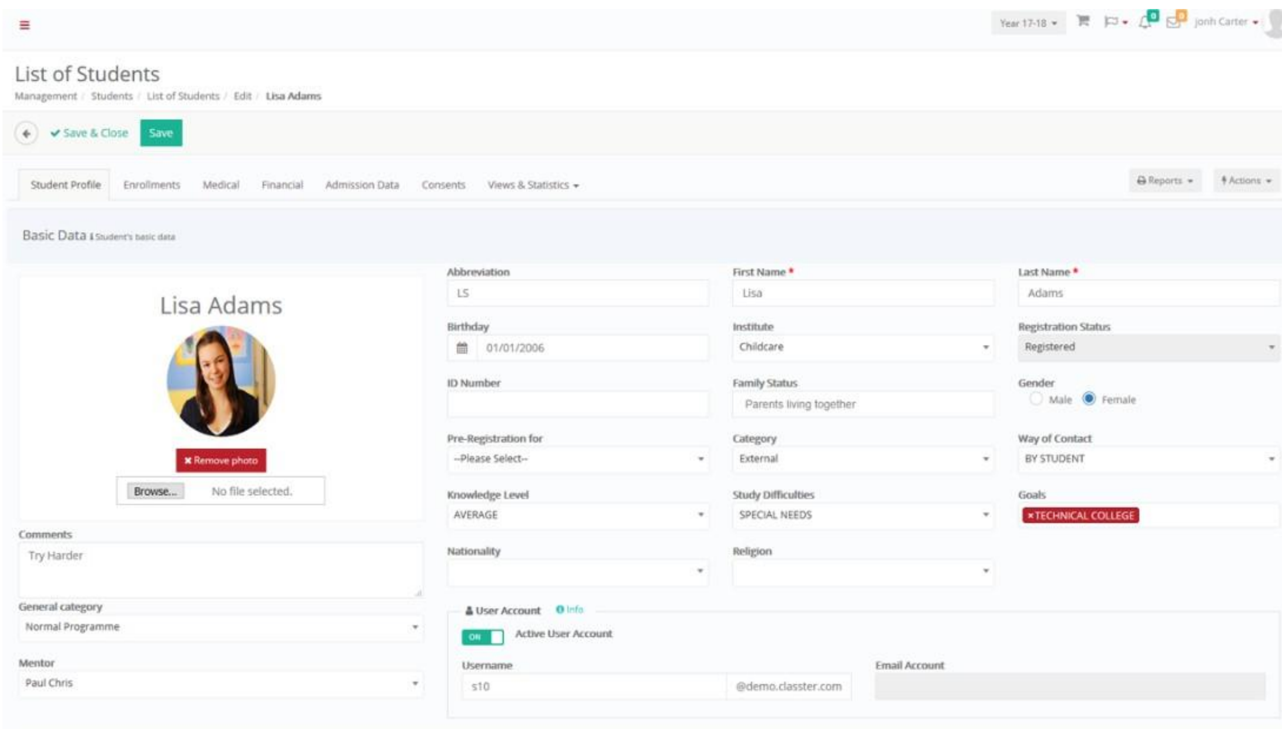
The system should be work properly in different platforms with less or no modifications.

2.7 Literature Review

Similar systems in the studying domain are needed to identified to get a vast understanding about the proposed one. It will help to identify basic fundamentals and possible drawbacks of the system. It will be helped to create a best system for the problematic domain.

2.7.1 Classter

Classter website supplies an all-in-one, cloud-based platform that provides an affordable solution for the administration of educational institutions. It provides different type of solutions such as LMS, ERP, CRM etc.



2.7.1 Classter User Interface

Functional features of Classter as below,

- Educational Programs & School Registration Management

- Contacts Detailed Cards & Confidential Info
- Extra-Curricular Activities Management
- Subjects & Groups Management
- Timetable Management
- Document Management

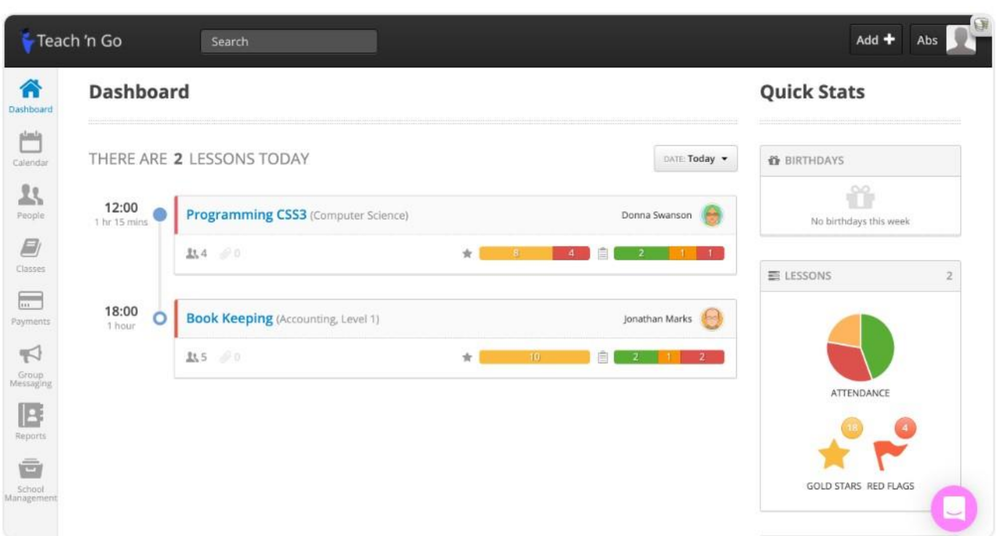
Non-Functional features of Classter as below,

- User Friendly Interface
- Easy to study

2.7.2 Teach 'n Go

Teach 'n Go is a fully integrated school administration software packed with a lot of benefits to meet any schools' requirements.

Following figure 2.7.2 shows a screenshot of Teach 'n Go web-based interface.



2.7.2 screenshot of Teach' nGo

Functional features of Teach 'n Go as below,

- Class scheduling
- Engage students
- Manage payments
- Manage school branches

Non-Functional features of Teach 'n Go as below,

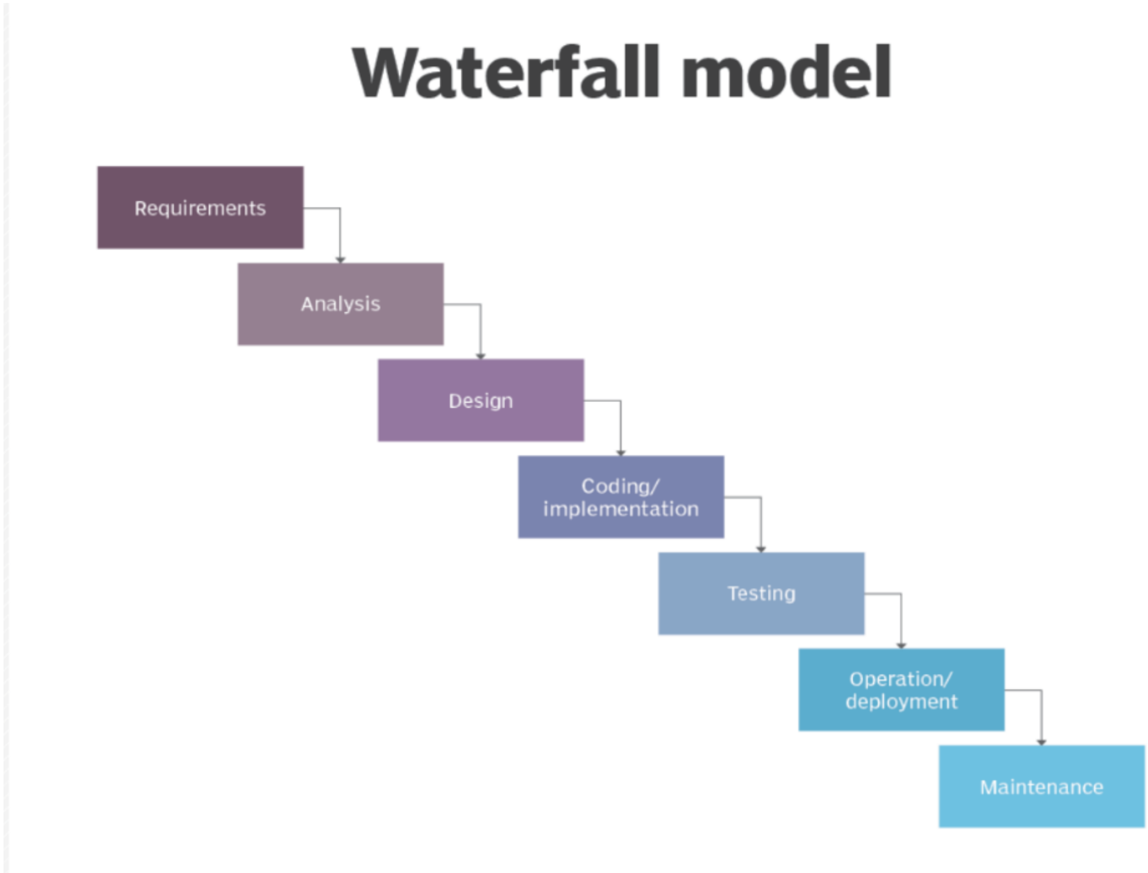
- User Friendly Interface
- Professional Support from development team

2.8 Process Model

An abstraction of the software development scenario is process model. The process models specify the stages and order of the process. Especially, a representation of the order of stages in the development process is called as a process model.

Even though the waterfall model is the most famous process model, it is not suitable if the requirements aren't clear. Incremental and extreme models are combined by the evolutionary prototyping. If the project's requirements aren't clear this model is suitable. Rapid Application Development (RAD) is established on parallel developed components or functions based iterative model and prototyping.

The following figure 2.8.1 is shown the structure of the waterfall model.



2.8.1 Waterfall method

For the development of ECMS, the waterfall model was chosen as the process model. Following advantages can be specified for the selection of waterfall model.

- Community support can be totally accessible because of the widely used technique.
- Easy and simple to understand and utilize.
- Each phase has a specific review process and deliverables.

Apart from this, some features of other modules such as, prototyping is used with waterfall model as needed.

Chapter 3 | Design

3.1 Introduction

The best solutions for the crucial problems, which are identified in the analyzing phase are made by the design phase of the system development process. The completed sketch of the proposed system is setup at the end of the design phase.

The selected development strategy with explanations, used design patterns and techniques and some scraps of design of the Education Center management are described by this chapter.

3.2 Development Strategy

Software development procedures are the divergent options available in translating the requirements into a working system. When considering about the development of ECMS, different strategies are used for it.

Purchasing an off-the shelf software can be minimized some kind of development cost and time. At some cases, it will not completely tally with the needed requirements may have included some other unnecessary features which makes extra cost and excess impediment to the system.

When comparing with distributed systems, Standalone systems are minimized some security susceptibilities. But standalone systems are unable to make easier for availability and remote accessing. As the excessive usage of the mobile devices such as smart phone mobile app facility can be another suitable option. However, it will not be better as a complete answer when referring a wide range system which require more capacity power.

As the development strategy for ECMS, the development of a web-based system from scratch was selected. Remote accessing with 24 x 7 consisted availability is provided by web-based system, as well as the best answers for the identified drawbacks are made by developing from scratch without difficulties.

3. 3 Design Techniques

System designing and modeling are based on design techniques which are well defined perspectives. When choosing a design technique, the designer must be paid more attention for the best one because, each of every pattern have their own features, advantages and disadvantages.

The system is adjusted of many little units knowing as functions in the structured design. Structured design technique is used by various diagrams such as data flow diagram is defined a flowing of data through the system, structure chart represents the system with its lowest manageable situation by breaking down of it and context diagrams are shown the margin between the system.

The most famous system design approach is object-oriented design which allows the software implementation based on the theme of objects. Classes, objects, encapsulation, inheritance and generalization for the design which are the concept of object orientation, used by OOD. Unified Modeling Language is a dominant modelling language which is provided a multi collection of diagrams with different purposed for system modeling. OOD is integrated with UML diagrams well. Object Oriented Programming is used in the implementation phase directly which was allowed by OOD.

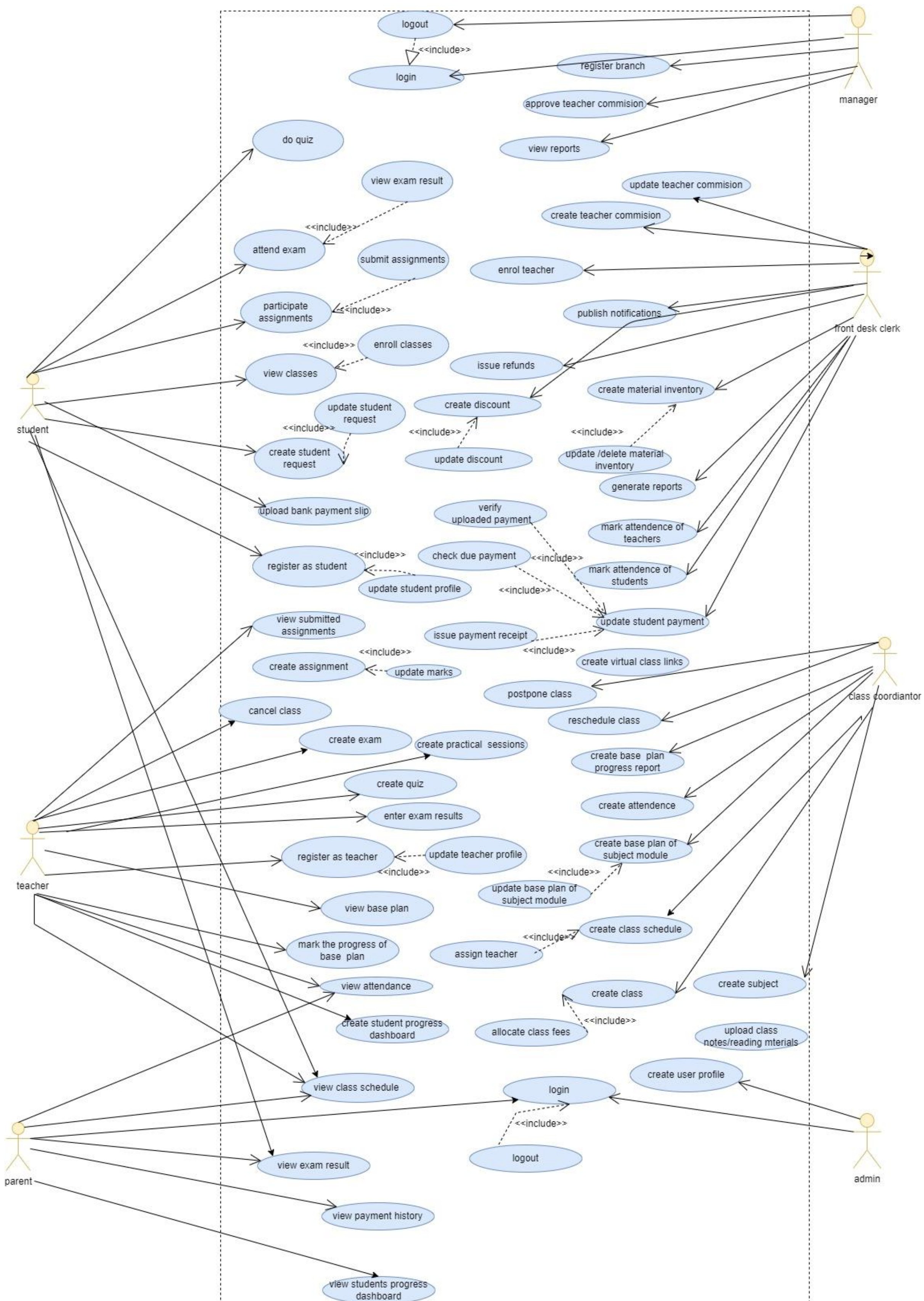
As the design technique of the ECMS, OOD with UML was selected due to the special features it has and the advantages that are explained earlier. Use case diagrams, sequence diagrams, activity diagrams and class diagrams were applied to identify the relationships between the system and its actors, represent the flow of activities in a particular way, show the relationship logic between the objects within the system in time process and to represent the relationships between the objects appropriately.

3.4 Design of ECMS

3.4.1 Use Case Diagram of ECMS

Use case diagrams are showed the interaction of users with system. It is provided clear idea about the system.

The overall use case diagram for the proposed system is shown below in figure 3.4



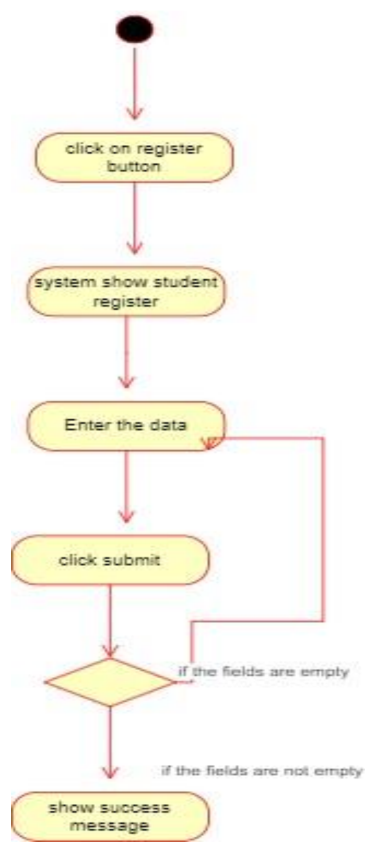
logout
<<include>>
login
register branch
approve teacher commision
view reports
manager
do quiz
view exam result
update teacher commision
create teacher commision
<<include>>
attend exam
submit assignments
enrol teacher
participate assignments
<<include>>
publish notifications
front desk clerk
<<include>>
enroll classes
view classes
issue refunds
create material inventory
update student request
create discount
student
<<include>>
<<include>>
<<include>>
create student request
update discount
update /delete material inventory
generate reports
upload bank payment slip
verify uploaded payment
mark attendence of teachers
<<include>>
check due payment
<<include>>
register as student
mark attendence of students
update student profile
<<include>>
update student payment
view submitted assignments
issue payment receipt
<<include>>
<<include>>
create virtual class links
create assignment
update marks
postpone class
class coordiantor
cancel class
reschedule class
create exam
create practical sessions
create base plan progress report
create quiz
create attendence
enter exam results
create base plan of subject module
register as teacher
update teacher profile
<<include>>
<<include>>
teacher
update base plan of subject module
view base plan
create class schedule
mark the progress of base plan
<<include>>
assign teacher
view attendance
create class
create subject
create student progress dashboard
allocate class fees
<<include>>
upload class notes/reading mterials
create user profile
view class schedule
login
<<include>>
logout
parent
view exam result
admin
view payment history
view students progress dashboard

3.4 use case diagram for proposed system

3.4.2 Activity Diagram for the proposed system

Activity diagram shows dynamic behavior of the system. It describes the flow from one activity to another activity.

Following figure 3.4.2 shows the Activity Diagram for Register Student.



3.4.2 Activity Diagram for Register Student

3.4.3 Use Case Narratives

They are explained how the interaction takes place step by step between the user and system.

Following Table 3.4.3.1 shows the Use Case Narrative for Register Student.

Use Case No	**01**	
Name	**Register Student**	
Description	**Register a new student into the system.**	
Precondition	• The student must provide necessary information for registration.	
Post condition	• The student is successfully registered in the system with a unique RegNo. • students can access the student's information for future reference.	
Basic Flow	**User response**	**System Response**
	01.click the register button	02.Show registration form
	03.Enter relevant details (First Name, Last Name, Email Address Line 1, Address Line 2, Address Line 3, Tel No (Home), Mobile No, DOB, Gender, Grade, Subject, District, Username, Password, confirm password)	04.Validate the data (first name is required, last name is required, invalid email address, email is already existing, DOB is required, Grade is required, Subject is required, password must be at least 8 characters, password is not matched. Recheck)
		05.Store data to the database
		06.Show success message and reg no
Alternative Flow		

Table: 3.4.3.1 Use Case Narrative for Register Student

Following Table 3.4.3.2 shows the Use Case Narrative for create assignment.

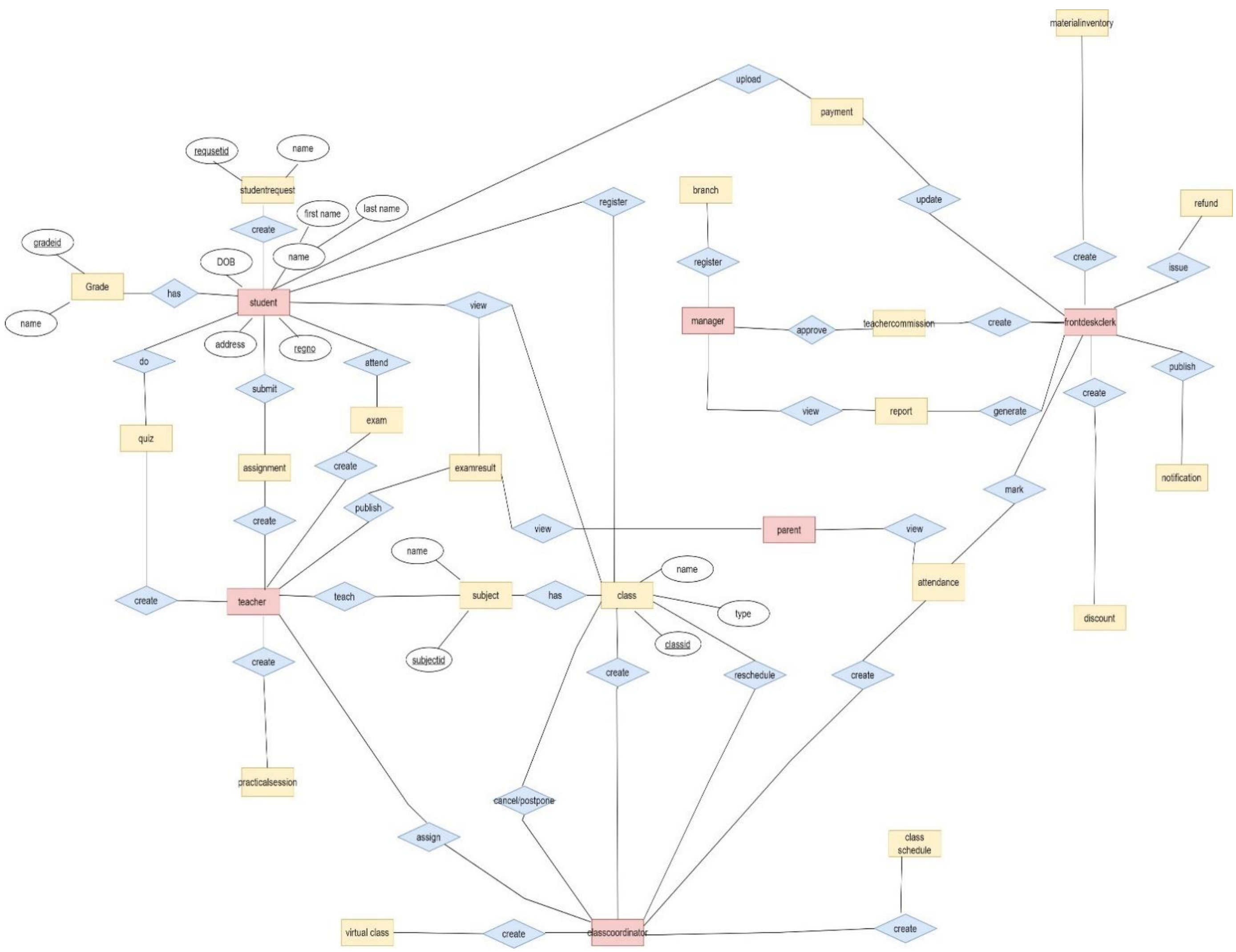
<table>
<tr><td>Use Case No</td><td colspan="2">02</td></tr>
<tr><td>Name</td><td colspan="2">Create Assignment</td></tr>
<tr><td>Description</td><td colspan="2">Teacher create a new assignment in Assignment management module.</td></tr>
<tr><td>Precondition</td><td colspan="2">The teacher must be logged into the Assignment management module.</td></tr>
<tr><td>Post condition</td><td colspan="2">The assignment is successfully created and is accessible to the specified class
Students enrolled in the class can view the newly created assignment and submit their work before the due date.</td></tr>
<tr><td rowspan="6">Basic Flow</td><td>User response</td><td>System Response</td></tr>
<tr><td>01.click the create assignment section.</td><td>02.show required list
Assignment title
Due date and time
Reading material</td></tr>
<tr><td>03.teacher enter the details (Assignment title, Due date and time, Reading material)</td><td>04.Validate the data (Due date must be required)</td></tr>
<tr><td>05. click create assignment button.</td><td>05.successfully created the assignment message</td></tr>
<tr><td></td><td>06.use case exist.</td></tr>
<tr><td>Alternative Flow</td><td colspan="2">If the teacher decides to cancel the assignment creation process at any point, he can cancel the operation.</td></tr>
</table>

Table 3.4.3.2 Use Case Narattive for Create Assignment

3.5 Data modeling

Database design is deciding what data must be stored and how the data interrelated. Entity Relationship (ER) Diagrams are data model diagrams which helps to identify logical structure of database.

The ER Diagram of the proposed system is shown below in figure 3.5



3.5 ER Diagram for proposed system

3.6 User interface Design

User interface is one of the crucial features in evaluating user friendliness of the ECMS. It is the main component the user sees and interacts with.

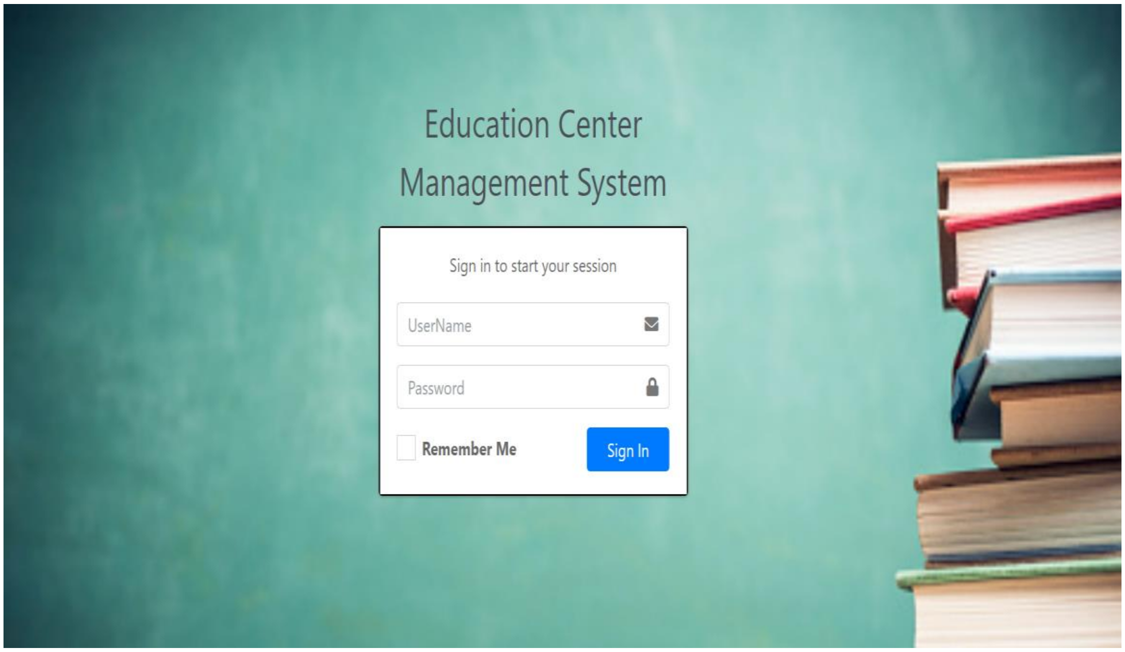
Some of considerations that use for user interface design were listed below.

- Simplicity of presenting information
- Understandable graphics and pictures
- Provide clear feedbacks for user actions

3.6.1 System Login Form

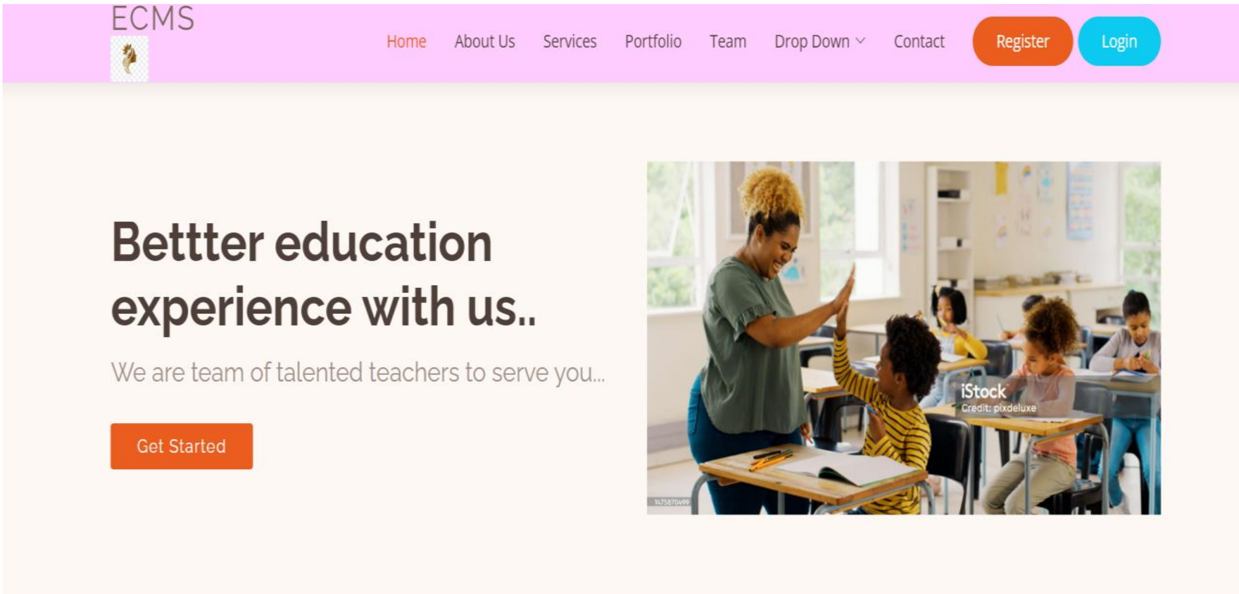
System users can login to system using the login form. Website navigation bar contains login link and it opens the login form.

The interface of login form is shown below in figure 3.6.1



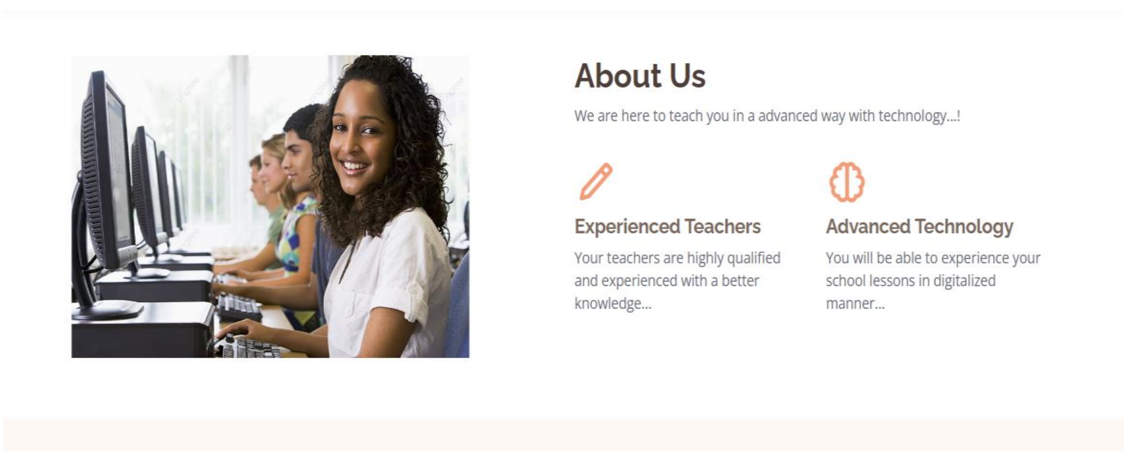
3.6.1 System login interface

The interface of web interface is shown below in figure 3.6.2



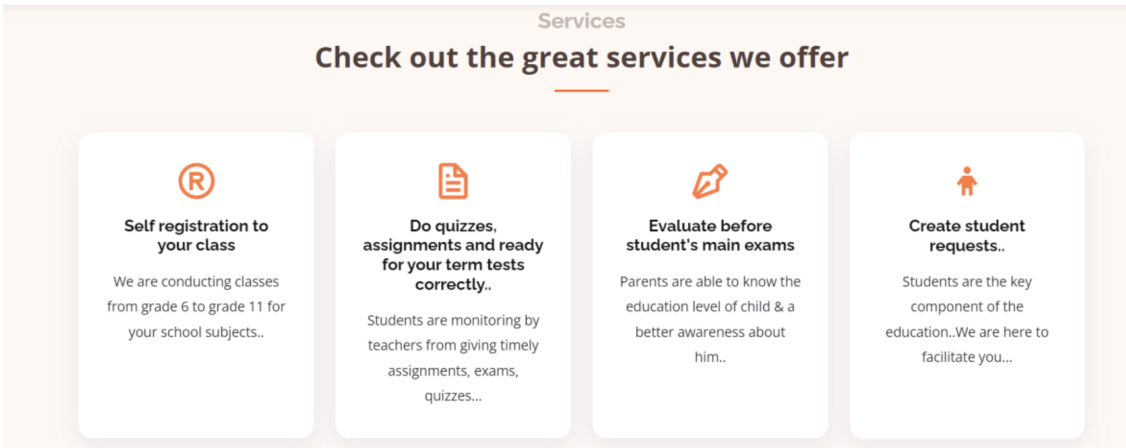
3.6.2 Web Interface

The interface of About Us is shown below in figure 3.6.3



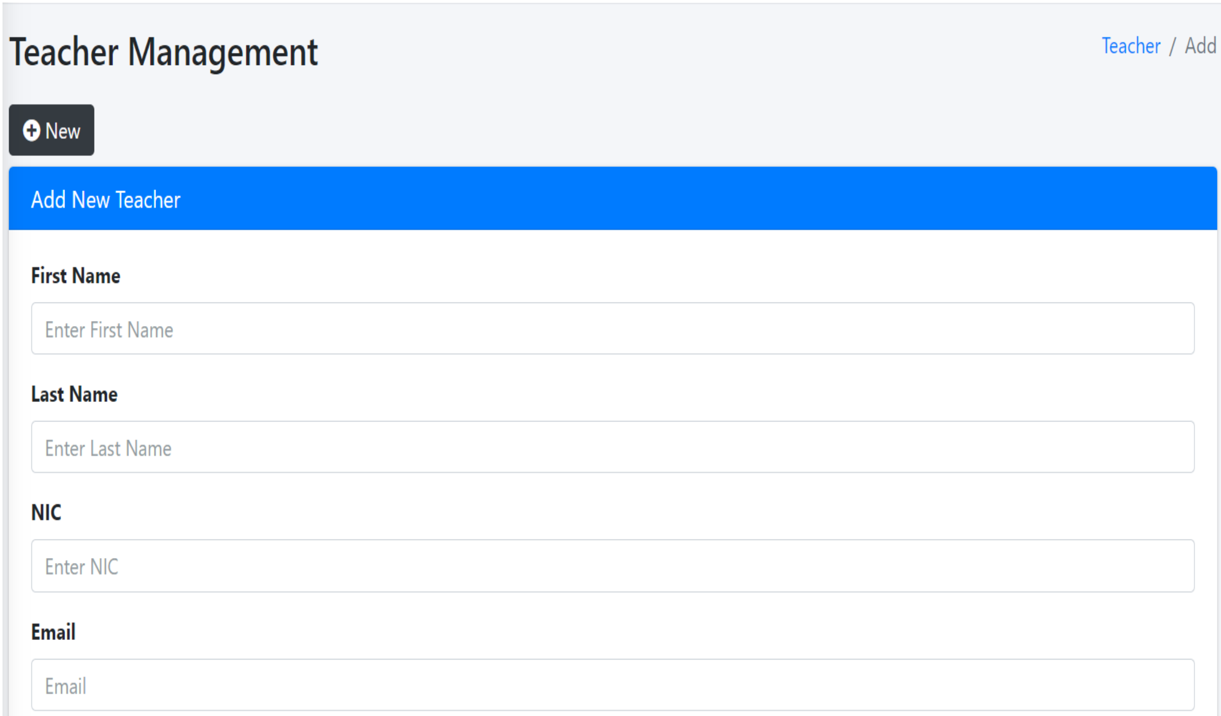
3.6.3 About Us in Web interface

The interface of Services Offer is shown below in figure 3.6.4



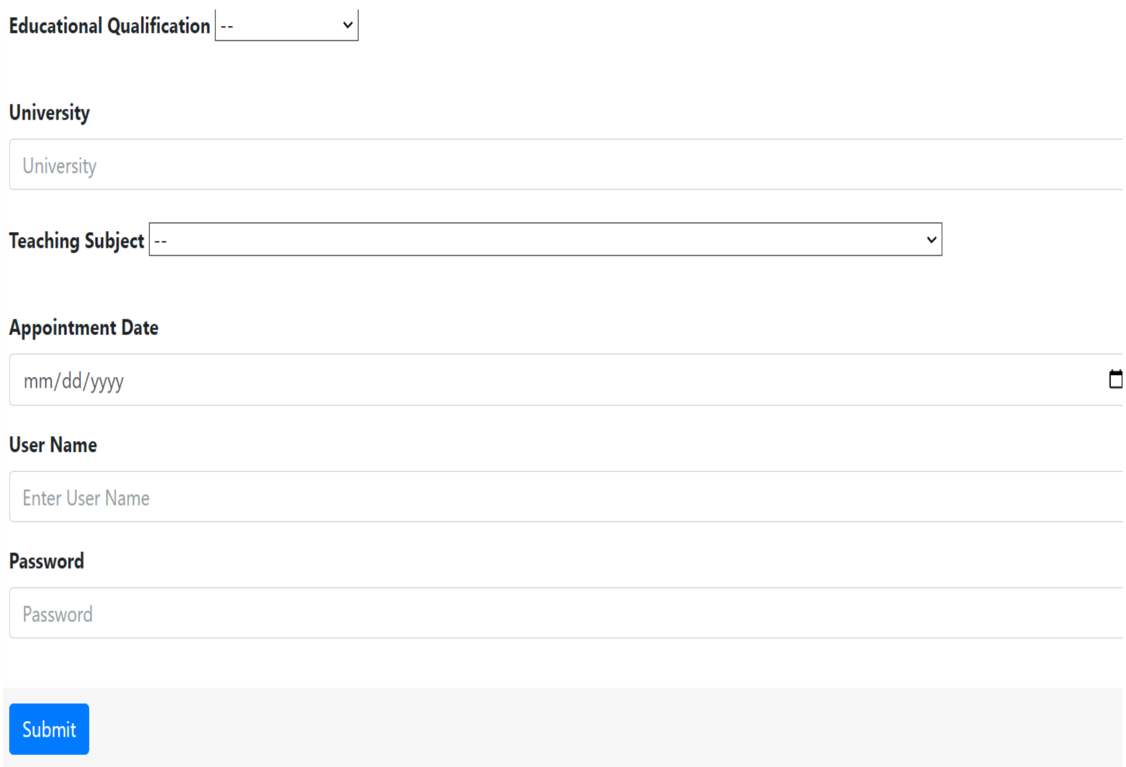
3.6.4 Services offer in Web Interface

The interface of Add Teacher is shown below in figure 3.6.5



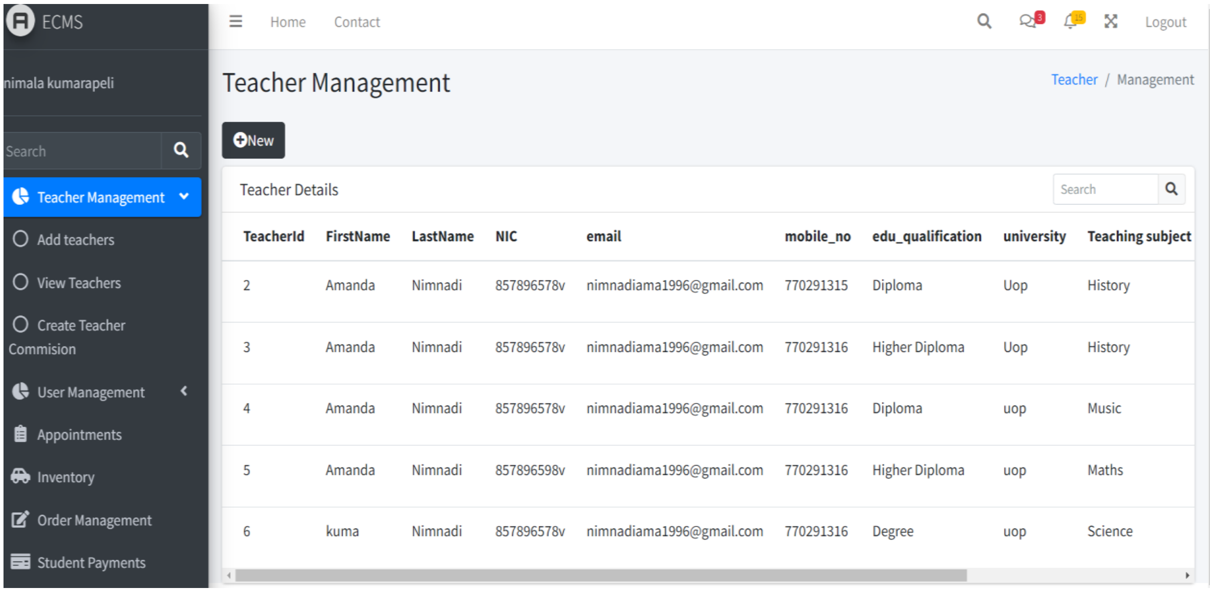
3.6.5 Add Teacher in Web Interface

The interface of Add Teacher is shown below in figure 3.6.5



3.6.6 Add Teacher in Web Interface

The interface of Teacher Details is shown below in figure 3.6.7



3.6.7 Teacher Details in Web Interface

Chapter4| Implementation

4.1 Introduction

In the implementation phase the design which made in the design phase is developed. Selecting programming languages, other tools such as framework selecting hardware platform, coding the system are occurred during this phase.

This chapter explains implementation framework, PHP framework, the tools used for development, structure of the application and the major code segments explanation.

4.2 Implementation environment

Implementation environment of the ECMS is given in the following table 4.1

Hardware	Software
Intel core i5 processor	Windows 10 - 64 bit
	XAMP server • Apache 2.4 • PHP 7.0 • PHPmyadmin
8 GB RAM	MySQL server 8.0
Internet connection	Apache NetBeans IDE 12.4
Monitor	Microsoft office 365
512 GB SSD hard disk	Chrome and Firefox

table 4.1 Implementation environment of the ECMS

4.2.1 Development tools and techniques

Client-side data manipulation is done by server-side programming language such as PHP, Java, Nodejs etc. The framework using is important because it gives the chance to developers to give more attention to the business process more than the coding. As well as developer does not worry about the efficiency of the whole system because the framework is concerned about it. It is reduced the repetitive codes that we are using.

- HTML

  HTML (Hypertext Markup Language) is used for the basic structure of the webpages.

- CSS

  CSS is used for defining the styles and the layout of the web pages.

- JavaScript

  It is used to apply the dynamic situations of the static web page.

- Jquery

  DOM manipulation and inbuilt functions are enhanced by JavaScript development productivity which is enabled by JavaScript library.

- Ajax

  It helps to increase the performance of project and build single page application for some modules.

- JSON

  It is applied for passing data to a one platform to another platform

- PHP

  It is used as the backend programming language. PHP is an open source language.

Database Management Sytem

DBMS software is used to write, change, retrieve and delete data from a database.

- MySQL Workbench

  It is used for visible indicating of database. MySQL community edition can be downloaded freely. Community edition is important for managing data smaller to larger businesses though they had an enterprise edition with so many functionalities.

4.2.2 Reused Code Modules

- Bootstrap
  It is an open source toolkit for developing with HTML, CSS and JS. It is used for building responsive, mobile web projects with the world's most famous front-end component library.
- Full calendar
  It is important for displaying payment notification dates. It is up to the user to add this functionality through Full calendar's API.

- jQuery Smart Wizard
  It is a flexible and extremely customizable jQuery Smart Wizard plugin with the help of Bootstrap. It gives a neat and stylized interface for the forms, registration steps etc. and easy to implement.

4.4 Module structure

Education Center Management System is included both a website and a system which gives access to registered users. Some main modules of the system are student management module, payment management module, teacher management module, exam and assignment management module and report generation.

Major Code Segments

used major PHP codes in ECMS are presented in this section.

User Login

Login form validates username and password. By studying following code segment, it can be identified how user login to the system and how it generates a session.

```
<?php
session_start();
include '../function.php';
?>
<?php
        if ($_SERVER['REQUEST_METHOD'] == 'POST') {
           extract($_POST);
            $username = dataClean($username);
             $message = array();
             if (empty($username)) {
                            $message['username'] = "User Name should not be
empty";
                        }
                 if (empty($password)) {
                            $message['password'] = "password should not be empty";
                        }
                      if (empty($message)) {
```

```
            $pw = password_hash($password, PASSWORD_DEFAULT);
            $db = dbConn();
            $sql = "SELECT * FROM users u INNER JOIN employee e ON
e.UserId=u.UserId WHERE u.UserName='$username'";
            $result = $db->query($sql);
            if ($result-> num_rows == 1) {
              $row=$result->fetch_assoc();
               if (password_verify($password, $row['Password'])) {
              $_SESSION['USERID']=$row['UserId'] ;
              $_SESSION['FIRSTNAME']=$row['FirstName'] ;
               $_SESSION['LASTNAME']=$row['LastName'] ;
               header("Location:dashboard.php");
              } else {
              $message['password'] = "invlid user name or password!...";
          }

            }else{
                $message['password'] = "invlid user name or password!...";}
}
?>
```

Student Registration

Following code segment shows how student is registered for the institute. All input fields get validated and check if there are existing user accounts, if all validations get confirmed, the registration process is done.

```
<?php ob_start();session_start();

include 'header.php';include '../function.php';

include '../mail.php';

?>

<main id="main"><!-- ======= Contact Us Section ======= -->  <section
id="contact" class="contact">  <div class="container" data-aos="fade-up"><div
class="section-title"> <h2>Student</h2><p>Register</p></div><div class="row
justify-content-center <div class="col-lg-7 mt-5 mt-lg-0 d-flex align-items-stretch "
data-aos="fade-up" data-aos-delay="200">

 <?phpif ($_SERVER['REQUEST_METHOD'] == 'POST') {extract($_POST);

                $first_name = dataClean($first_name);  $last_name =
dataClean($last_name); $pnic = dataClean($pnic); $address_line1 =
dataClean($address_line1); $address_line2 = dataClean($address_line2);
$address_line3 = dataClean($address_line3);

$message = array();

                       //Required validation---------------------------------------  if
(empty($first_name)) {

                           $message['first_name'] = "The first name should not be blank...!";

                       }                       if (empty($last_name)) {

                           $message['last_name'] = "The last name should not be blank...!";

                       } if (empty($pnic)) {

                           $message['pnic'] = "Parent NIC is required ...!";

                       if (!isset($dob)) {

                           $message['dob'] = "Date Of Birth is required";

                       }
```

```
if (!isset($gender)) {
                        $message['gender'] = "Gender is required";
                    }                       if (!isset($grade)) {
                        $message['grade'] = "Grade is required";
                    }
                    if (!isset($subject)) {
                        $message['subject'] = "Subject is required";
                    }
                    if (empty($user_name)) {
                        $message['user_name'] = "User Name is required";
                    } if (empty($password)) {
                        $message['password'] = "Password is required";
                    }
  //Advance validation----------------------------------------------
                    if (ctype_alpha(str_replace(' ', '', $first_name)) === false) {
                        $message['first_name'] = "Only letters and white space
allowed"; }
                    if (ctype_alpha(str_replace(' ', '', $last_name)) === false) {
                        $message['last_name'] = "Only letters and white space
allowed";
                    }
                    if (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
                        $message['email'] = "Invalid Email Address...!";
                    } else {
                        $db = dbConn();
                        $sql = "SELECT * FROM students WHERE Email='$email'";
                        $result = $db->query($sql);
```

```
 if ($result->num_rows > 0) {    $message['user_name'] = "This User Name already exsist...!";                          }

 } if (!empty($password)) {if (strlen($password) < 8) {

                          $message['password'] = "The password should be 8 characters more";

                      }                     }if ($password !== $cpassword) {

                      $message['cpassword'] = "Password is not matched..Recheck ";

                  if (empty($message)) { //Use bcrypt hashing algorithem

                      $pw = password_hash($password, PASSWORD_DEFAULT);

                      $db = dbConn();

                      $token = bin2hex(random_bytes(16));

                      $sql = "INSERT INTO `users`(`UserName`, `Password`,`token`,`is_verified`) VALUES ('$user_name','$pw','$token','0')";

                      $db->query($sql);$user_id = $db->insert_id;$reg_number = date('Y') . date('m') . date('d') . $user_id;

                      $_SESSION['RNO'] = $reg_number; $sql = "INSERT INTO `students`(`FirstName`, `LastName`, `Email`,`pnic`, `AddressLine1`, `AddressLine2`, `AddressLine3`, `TelNo`, `MobileNo`,`DOB`,`Gender`,`GradeId`,`SubjectId`,`DistrictId`,`RegNo`,`UserId`) VALUES ('$first_name','$last_name','$email','$pnic','$address_line1','$address_line2','$address_line3','$telno','$mobile_no','$dob','$gender','$grade','$subject','$district','$reg_number','$user_id')"  $db->query($sql)$msg = "<h1>SUCCESS</h1>   $msg .= "<h2>congratss...</h2>"; $msg .= "<p>your account has been successfully created..</p> $msg .= "<a href='http://localhost/ecmstest/verify.php?token=$token'>click here to verify your account</a>";  sendEmail($email, $first_name, "Account verification", $msg);

                      header("Location:register_success.php");

                  } }

              ?>
```

Chapter5| Evaluation

5.1 Introduction

Software evaluation is the process of measuring the performance and utility of software assets in businesses' inventories. The purpose of software evaluation is to defect dysfunctional and unused software. This consists the preliminary development of software, its maintenance and updates, till needed software product is developed which satisfies the expected requirements.

Evaluation phase in the application development is the phase that confirms the components and modules with requirements and ensures the system working worthily with the help of that cases. Validation and verification process were done to identify whether system meets specifications and provides it's intended purpose.

Verification will be helped for identify the developed software is high quality which is meant by the correctness of system. Unit testing, integration testing, automated testing is done. Validation will be helped identify whether the system fulfills client's requirements. Usability and customer acceptance testing is done.

5.2 Testing Approach

There are many testing approaches to confirm the system step by step. Each step verifies the one specific aspect of the system and all the strategies interconnected with each other.

Following a short introduction about testing approaches used for the project.

Unit Testing

It is the way of testing a unit – the smallest piece of code that can be logically isolated in a system.

Integration Testing

The aim of integration testing is to test the interfaces between modules and expose any defects that may arise when these components are integrated and need to interact with each other.

Automated Testing

A specialized program is automated testing. Developer has to have vast knowledge about hoe to use such tool.

Usability Testing

It is done by end-users to enhance overall user experience when using the system.

Acceptance Testing

This testing is done after the development of the project. It is done to determine whether the system meets the requirements of clients.

5.3 Test Plan

It is included the testing, scope, steps and activities for the system. SBM was tested using with the help of dynamic testing techniques (Unit testing, Functional testing, Acceptance testing) to verify that the system is free of defects.

5.4 Test Cases

Test cases and test results of login activity of ECMS

Following table 5.1 shows test cases and results of login activity of **ECMS**

No	Action	Expected Result	Result
1	Empty username and password	Login error	Expected result
2	Enter only a username	Login error	Expected result
3	Enter only a password	Login error	Expected result
4	Enter invalid username and password	Login error	Expected result
5	Enter valid username and invalid password	Login error	Expected result
6	Enter invalid username and valid password	Login error	Expected result

7	Enter valid username and password	Logged in to the system and navigate to dashboard	Expected result

Table 5.1: Test cases and results of login activity of ECMS

Test cases and test results of 'add student' activity of ECMS

Following table 5.2 shows test cases and results of 'add student' activity of ECMS

No	Action	Expected Result	Result/Status
1	Submit form with empty fields	Error message	Expected result
2	Enter invalid name like 'wimal008'	Validation error displayed	Expected result
3	Enter invalid nic like '967100471'	Validation error displayed	Expected result
4	Enter NIC that already exists in the system	Validation error displayed	Expected result
5	Enter email that already exist in the system	Validation error displayed	Expected result
6	Enter correct member details	Confirmation dialog box pops up	Expected result

*Table **5.**2 Test cases and results of add student activity of ECMS*

Test cases and test results of 'change student details' activity of ECMS

Following table 5.3 shows test cases and results of 'change student details' activity of ECMS

No	Action	Expected Result	Result/Status
1	Submit without making any change	Update button is disabled	Expected result
2	update invalid details such as name like 'wimala123'	Error message	Expected result
3	Update NIC as existing NIC in the system	Error message	Expected result
4	Update email as existing email in the system	Error message	Expected result
5	Make valid and correct changes and submit	Error message	Expected result

Table 5.3: Test cases and results of 'change student details' activity of ECMS

5.5 User evaluation

User evaluation was done while acceptance testing. A user evaluation form was handed out for few users of the proposed system. They were guided to fill out the form while using the system.

Following figure 5.1 shows the user feedback form

Education Center Management System User Feedback Form

Please mention your satisfaction level as follows;

1	2	3	4	5
Highly Dissatisfied	Dissatisfied	Average	Satisfied	Highly Satisfied

Functional and non-functional requirements	1	2	3	4	5
Accuracy of information					
UI Design					
Simplicity of functions					
Speed of response					
Speed of data retrieval					
Easiness of navigation					
Usefulness of generated reports					

Figure 5.1: User feedback form

6.Conclusion

Following chapter describes a summary of main chapters overall.

6.1Critical Evaluation

Before implementing the ECMS the Thaksala Education Center already used excel sheets and large ledger books to store the data. Old system was not an enhanced featured system for Thaksala Education Center. These vulnerabilities were solved by the new system.

In the analysis phase requirements were collected. A few meetings were conducted to gather user requirements. After that client requirements were studied carefully. Then in the design phase, answers for the requirements were made for the new project. UML diagrams were drawn for modeling. In the implementation process, codes were written according to the design. New proposed system was able to fulfil the requirements of the client.

- Efficiency
  ECMS system is able to maintain students and teachers details in an efficient manner. Because all students' details are valuable for statistical information of the education center.

- Accuracy
  While managing the low-income beneficiaries, it is important to assure the accuracy of the information.
  Then, new decision can be made in easily and reliable manner.

- Accessibility and Availability

  There are different types of users using the ECMS. They are in different managerial levels. Middle level management people like Front desk clerk, teacher, class coordinator use the system more frequently than higher management. They are able to log in to the system regularly and update their working. ECMS system ensure the availability and accessibility of the system.

6.2 Lessons learned

It was great experience to realize the business of the education center. Some practical issues were identified while developing the system. Various lessons were learned while developing ECMS.

Practical application

- Organized and used technical skills to realize the big picture of the ECMS.
- Realized the process that the theoretical knowledge translated into Software product.

Problem Solving Skills

- How to use the skills and knowledge to help in the work with a productive manner and improve the working efficiency.
- Got a vast knowledge while fixing errors, bugs and various issues while developing.

Professional Communication Improvement

- When dealing with the client.
- Demonstrate our system to the users.

Path to IT industry

- Realizing the business process in the client environment.
- Find the requirement from the user stories and create conceptual idea about the system feasibility.
- Proper planning and based on the project in a timely behavior.
- Software development tools and techniques such as Ajax, jQuery, Bootstrap, PHP, MySQL are experienced practically.

Work under a supervisor

- Got experience of working with an industry expert and used that instruction to minimize software development errors and bugs.
- Time and internet sources were used efficiently.

6.3 Future Improvements

- Introduce an online student payment for the students through a payment gateway
- Developing a mobile app for the users

Appendix A – System Manual

Hardware Requirements

Below mentioned hardware configurations are required to develop this system,

- At least Intel core i3 processor
- 8 GB Ram or higher
- At least 5GB free storage capacity
- Inkjet or laser printer (To print reports)

Software Requirements

Below mentioned software configurations are required to develop this system,

- Apache 2.4
- PHP 7.0
- PHPmyadmin
- MySQL server 8.0
- Any web browser
- Any Windows OS edition

  An internet connection is required to download necessary things for the system.

Program Installation

Below mentioned steps are required to install and run this program.

1. Install WAMP, XAMPP or a similar software to use as the Database Server

    1.1. Download link to XAMPP

    https://www.apachefriends.org/download.html

2. You need to install PHP, MySQL in your machine.

    2.1. Download link to PHP

    https://windows.php.net/download#php-8.2

3. Create a database using "PHPMyAdmin" tool in XAMPP or a similar tool that you've installed.

4. Extract to a directory. Here I'm using the same name as the zip folder "Education Center Management System".

5. Open the above folder in your preferred IDE.

6. Change the following variable values according to your local database server.

DB_HOST=127.0.0.1

DB_PORT=3306

DB_DATABASE=ecms_test

DB_USERNAME=root

DB_PASSWORD=

Note: Keeping the password field empty is not a problem, if db user has an empty password.

7. Now you can visit “localhost/ecms” and login using a given username and password.
Admin username: “nimala”
Password: 19960728

Appendix B –User Manual

User manual is a user guide document that uses to communicate and gives the help to end users who use the system. A user manual is presented whether the system is a software or hardware. The end users can be referred the system manual when they faced problems using the system.

Login Form

Login page of the system is shown by following figure B.1.This user interface is shown at the startup of the system. Login page will be restricted users who are able to access the system. Login credentials for each user are required to login to the system. Invalid username and password will be resulted a login error.

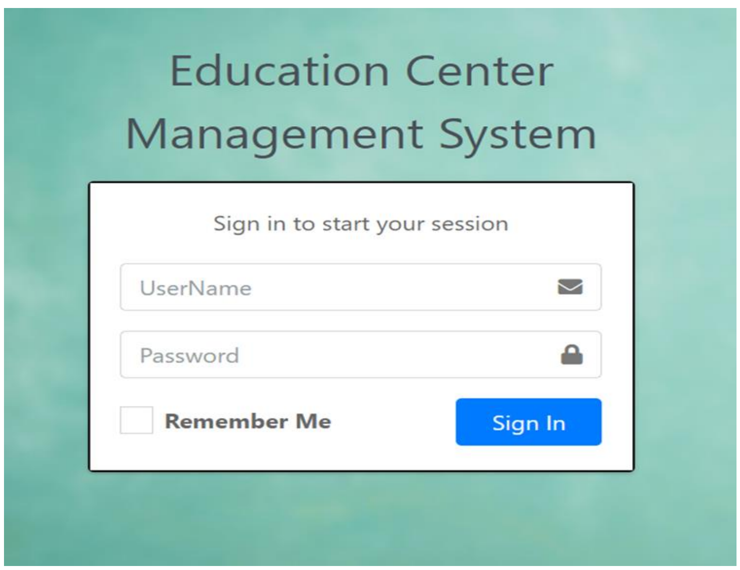


Figure B.1 ECMS login

Dash board of Student

Following figure B.2 shows the dashboard of Student of the system.

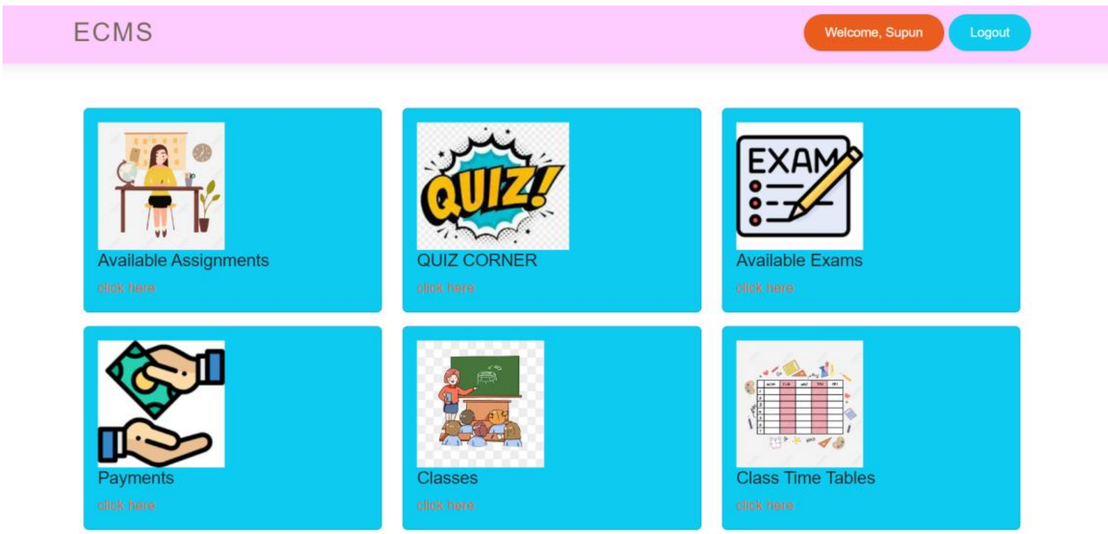


Figure B.2 Dashboard of Student

Main navigation bar of Front Desk Clerk dashboard

Following figure B.3 shows the dashboard window of a Front Desk Clerk of system,

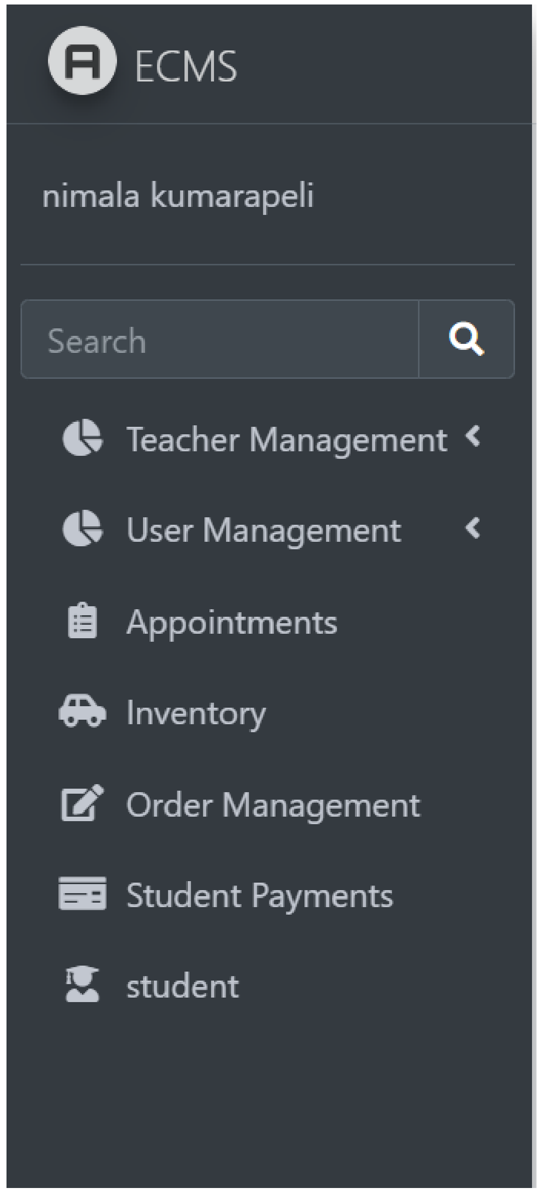


Figure B. 3 Main navigation bar of Front Desk Clerk dashboard

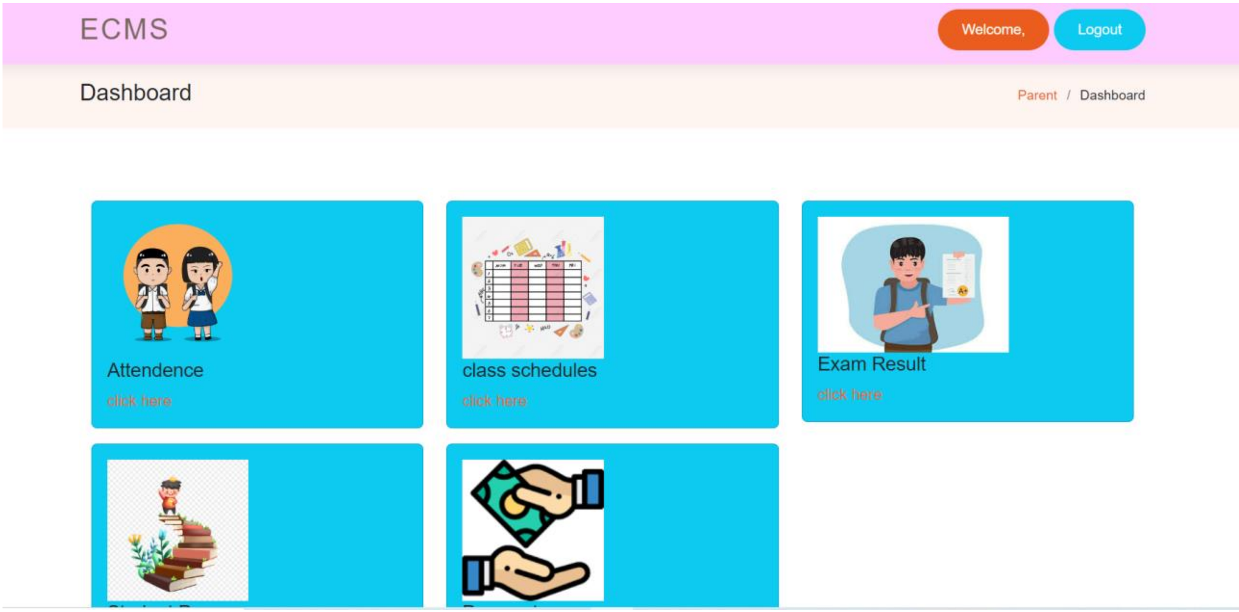


Figure B.4 Parent Dashboard view

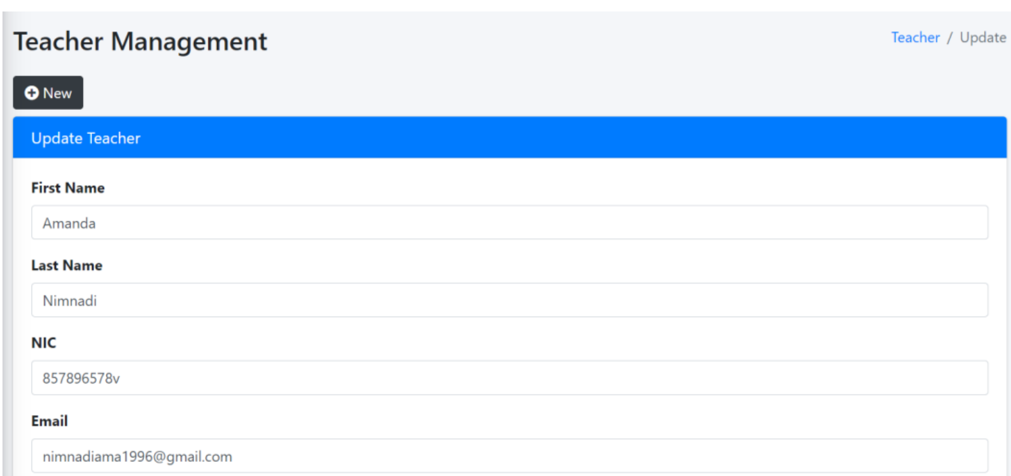


Figure B.5 Update Teacher view

Appendix C –Management Reports

Reports are used to make decisions in higher level management.

In ECMS following reports were identified according to the user requirements.

Following reports were identified according to user requirements,

- Reports on students
  - Attendance reports for each class
- Reports about classes
  - Student enrolment for each class
  - Outstanding payments for each class
- Reports about courses
  - Class reports related to subject
  - Revenue details according to course
- Reports about branches
  - Revenue details of each branch daily, weekly, monthly, yearly
  - Student registrations for each branch
  - Branch wise student’s outstanding payments
- Reports about payment
  - Daily, weekly. Monthly, yearly payments done by students
- Reports about refunds
  - Branch wise refunds for selected subjects
  - Subject wise refunds

Appendix D –Test Results

Test cases and test results of login activity of ECMS

Following table D.1 shows test cases and results of login activity of ECMS

No	Action	Expected Result	Result
1	Empty username and password	Login error	Expected result
2	Enter only a username	Login error	Expected result
3	Enter only a password	Login error	Expected result
4	Enter invalid username and password	Login error	Expected result
5	Enter valid username and invalid password	Login error	Expected result
6	Enter invalid username and valid password	Login error	Expected result
7	Enter valid username and password	Logged in to the system and navigate to dashboard	Expected result

Table 5.2: Test cases and results of login activity of ECMS

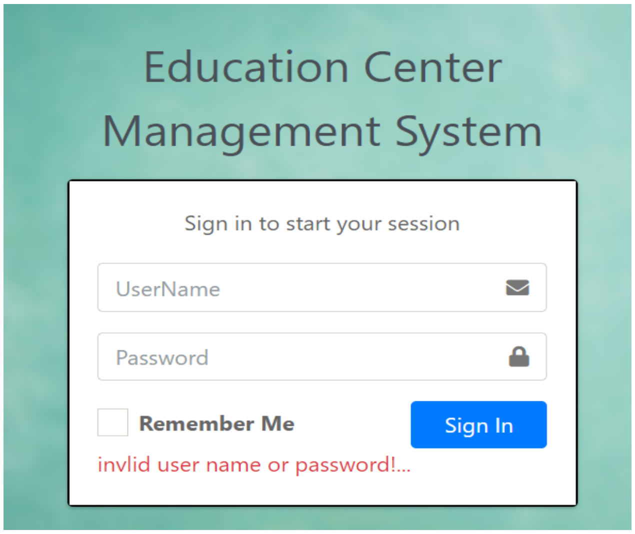


Figure D.1 Enter Invalid password in login activity

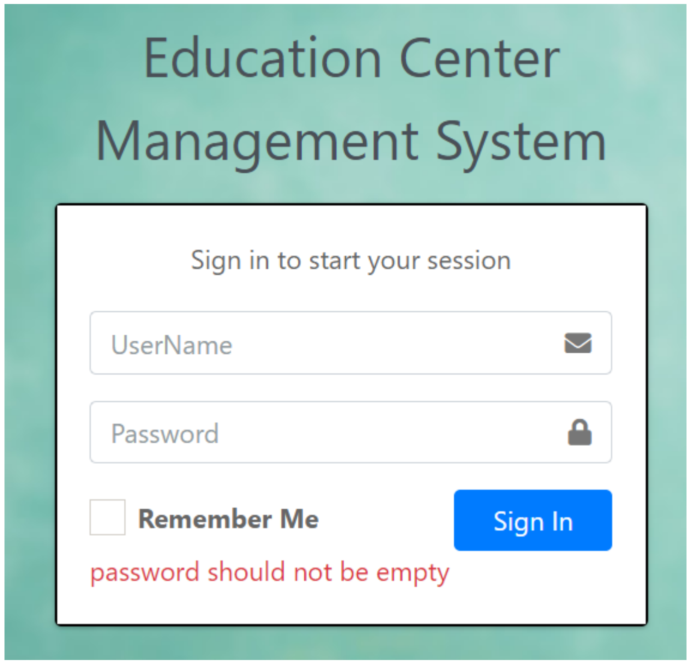
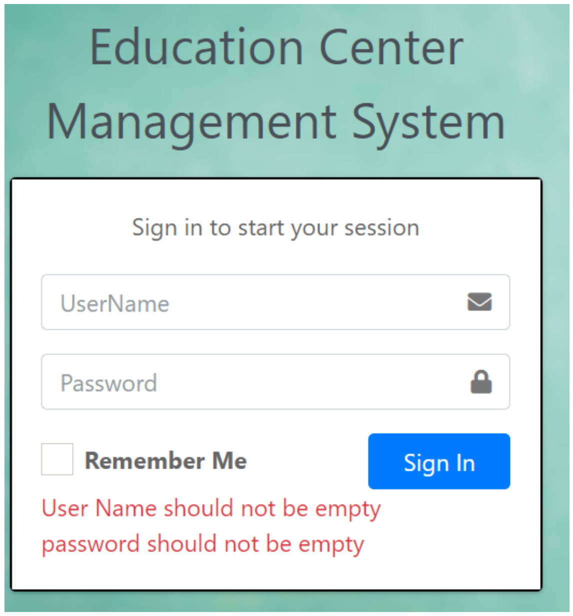
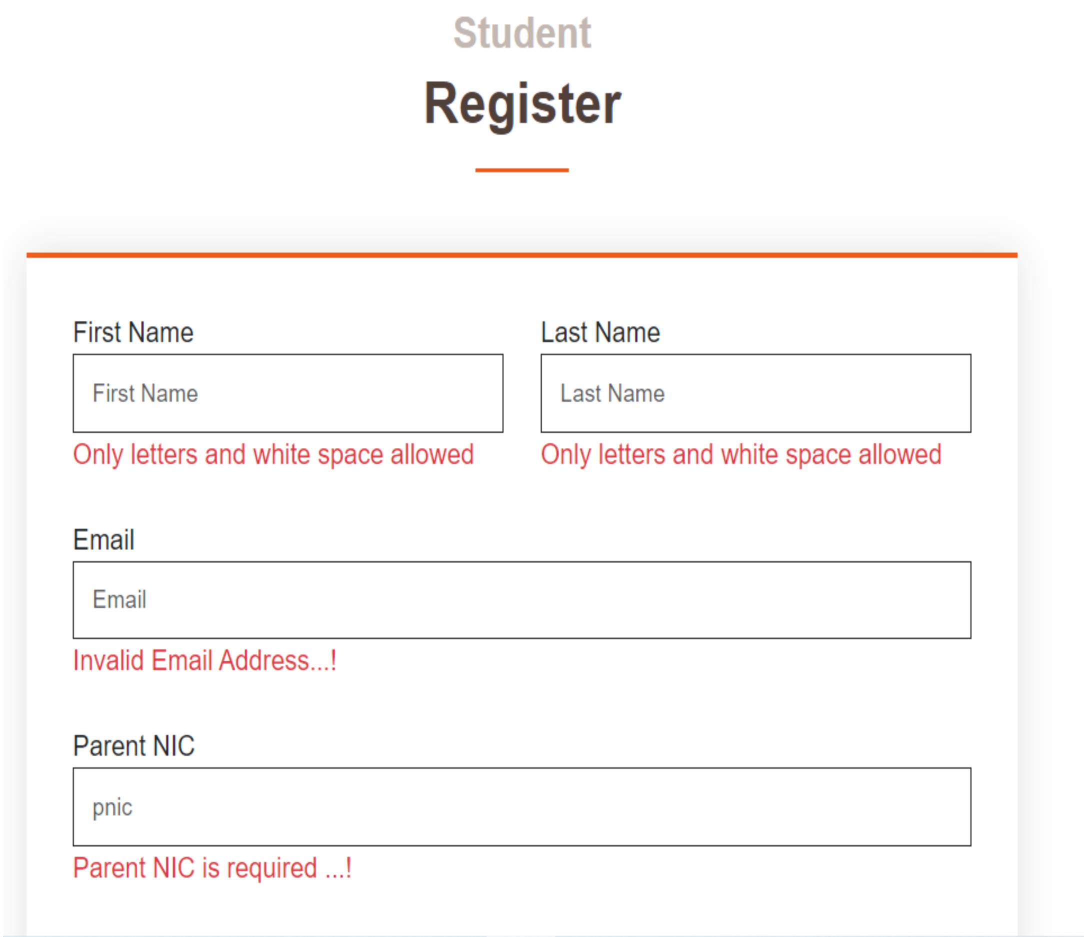


Figure D.2 Enter empty password in login activity

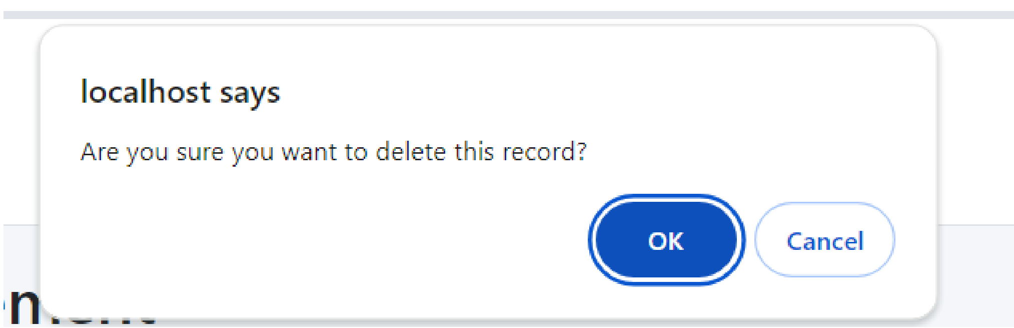
Figure D.3 Enter invalid username in login activity



D.4 Empty username and empty password in login activity



D.5 Validations of students add



D.6 Member delete confirmation view

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